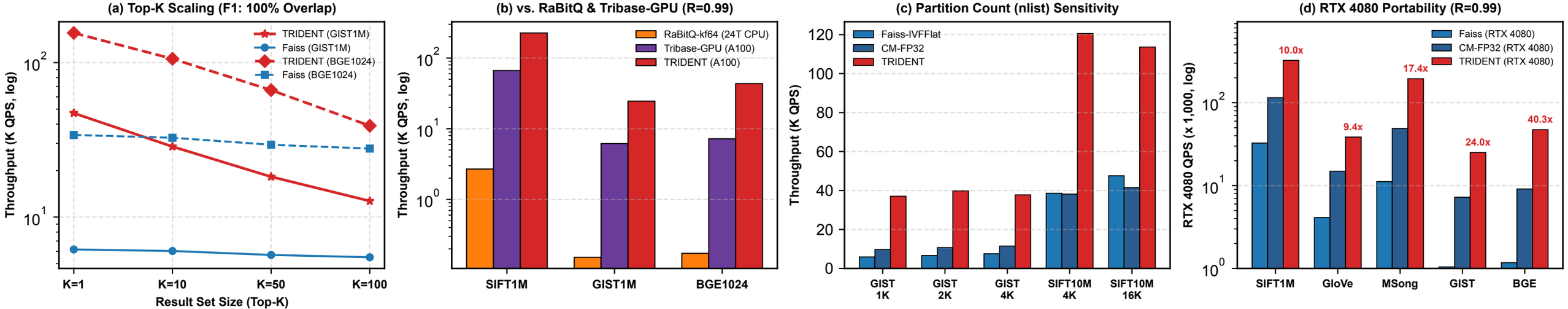
(a) Top-K Scaling (F1: 100% Overlap)
TRIDENT (GIST1M)
Faiss (GIST1M)
TRIDENT (BGE1024)
Faiss (BGE1024)
Throughput (K QPS, log)
10^2
10^1
K=1
K=10
K=50
K=100
Result Set Size (Top-K)
(b) vs. RaBitQ & Tribase-GPU (R=0.99)
RaBitQ-kf64 (24T CPU)
Tribase-GPU (A100)
TRIDENT (A100)
Throughput (K QPS, log)
10^2
10^1
10^0
SIFT1M
GIST1M
BGE1024
(c) Partition Count (nlist) Sensitivity
Faiss-IVFFlat
CM-FP32
TRIDENT
Throughput (K QPS)
120
100
80
60
40
20
0
GIST 1K
GIST 2K
GIST 4K
SIFT10M 4K
SIFT10M 16K
(d) RTX 4080 Portability (R=0.99)
Faiss (RTX 4080)
CM-FP32 (RTX 4080)
TRIDENT (RTX 4080)
RTX 4080 QPS (x 1,000, log)
10^2
10^1
10^0
10.0x
9.4x
17.4x
24.0x
40.3x
SIFT1M
GloVe
MSong
GIST
BGE